
EXERCISES

AI 504 activity: Canonical models

AI504 — Knowledge Representation

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1 Problem 1

Work over the set of nouns $P = \{p, q, r\}$. Let $\Gamma = \{\text{All } p \text{ are } q, \text{All } p \text{ are } r\}$. Compute each of the following three sets:

- $\{x \in P : \Gamma \vdash \text{All } x \text{ are } p\}$
- $\{x \in P : \Gamma \vdash \text{All } x \text{ are } q\}$
- $\{x \in P : \Gamma \vdash \text{All } x \text{ are } r\}$

1.1 Solution

Let set A,B,B be

$$A = \{\text{All } p \text{ are } p, \text{All } q \text{ are } p, \text{All } r \text{ are } p\}$$

The elements in P that satisfies this is $\{p\}$.

$$B = \{\text{All } p \text{ are } q, \text{All } q \text{ are } q, \text{All } r \text{ are } q\}$$

The elements in P that satisfies this is $\{p, q\}$.

$$C = \{\text{All } p \text{ are } r, \text{All } q \text{ are } r, \text{All } r \text{ are } r\}$$

The elements in P that satisfies this is $\{p, r\}$.

2 Problem 2

Now construct a P -model whose domain is also P , and where the interpretation of p, q, r are the three sets above respectively. (Caution: P plays two roles here: it is both the domain as well as the underlying signature. While this may be momentarily confusing, there is nothing wrong with it!) Draw your answer below.

2.1 Solution

$$\mathcal{M} = (P, \llbracket \cdot \rrbracket : P \rightarrow \mathcal{P}(P))$$

$$\llbracket p \rrbracket := \{p\}$$

$$\llbracket q \rrbracket := \{p, q\}$$

$$\llbracket r \rrbracket := \{p, r\}$$

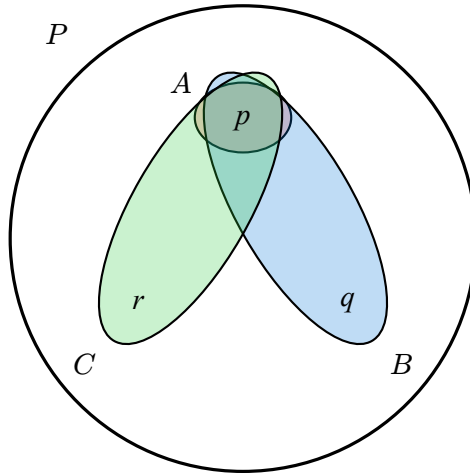


Figure 1: $\mathcal{M}$

3 Problem 3

Look at your model. What sentences does it satisfy? Write them all below. Does that set contain

3.1 Solution

We want to check if $\forall x, y \in \{p, q, r\} \mid \mathcal{M} \models \text{All } x \text{ are } y$

$\mathcal{M} \models \text{All } p \text{ are } p \quad \checkmark$

$\mathcal{M} \models \text{All } p \text{ are } q \quad \checkmark$

$\mathcal{M} \models \text{All } p \text{ are } r \quad \checkmark$

$\mathcal{M} \models \text{All } q \text{ are } p \quad \times$

$\mathcal{M} \models \text{All } q \text{ are } q \quad \checkmark$

$\mathcal{M} \models \text{All } q \text{ are } r \quad \times$

$\mathcal{M} \models \text{All } r \text{ are } p \quad \times$

$\mathcal{M} \models \text{All } r \text{ are } q \quad \times$

$\mathcal{M} \models \text{All } r \text{ are } r \quad \checkmark$

4 Problem 4

Now go exactly through the same process, but for the set $\Sigma = \{\text{All } p \text{ are } q, \text{All } r \text{ are } p\}$. Draw the resulting model below.

$\mathcal{M} = (P, \llbracket \cdot \rrbracket : P \rightarrow \mathcal{P}(P))$

$$\llbracket p \rrbracket := \{p, r\}$$

$$\llbracket q \rrbracket := \{p, q, r\}$$

$$\llbracket r \rrbracket := \{r\}$$

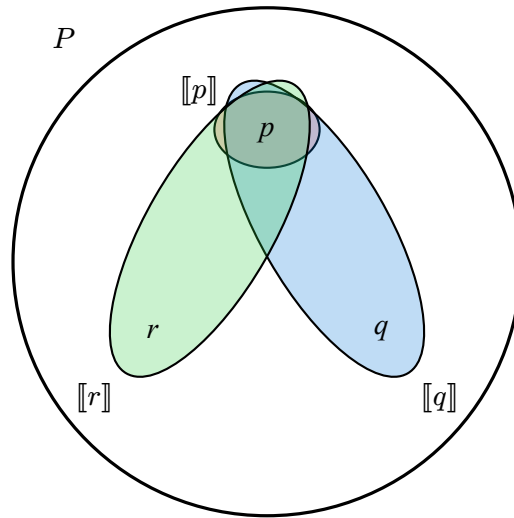


Figure 2: $\mathcal{M}$

5 Problem 5

Again, write the set of sentences that your model satisfies. Does it contain Σ as a subset?

5.1 Solution

$\mathcal{M} \models \text{All } p \text{ are } p \quad \checkmark$

$\mathcal{M} \models \text{All } p \text{ are } q \quad \checkmark$

$\mathcal{M} \models \text{All } p \text{ are } r \quad \times$

$\mathcal{M} \models \text{All } q \text{ are } p \quad \times$

$\mathcal{M} \models \text{All } q \text{ are } q \quad \checkmark$

$\mathcal{M} \models \text{All } q \text{ are } r \quad \times$

$\mathcal{M} \models \text{All } r \text{ are } p \quad \checkmark$

$\mathcal{M} \models \text{All } r \text{ are } q \quad \checkmark$

$\mathcal{M} \models \text{All } r \text{ are } r \quad \checkmark$

6 Problem 6

This construction is called the canonical model construction. It allows us to construct a model out of a set of sentences. Write another set of sentences below, and make it a little more complicated than the previous examples—maybe 3-5 sentences over a signature of 4-5 symbols.

6.1 Solution

Let $P = \{p, q, r, t\}$, and $\Gamma = \{(\text{All } p \text{ are } q), (\text{All } t \text{ are } r), (\text{All } r \text{ are } q)\}$

7 Problem 7

Draw the canonical model for the set of sentences you just created. Once you do this, check whether the model just created satisfies each one of those sentences.

7.1 Solution

$$\mathcal{M} = (P, \llbracket \cdot \rrbracket : P \rightarrow \mathcal{P}(P))$$

$$\llbracket p \rrbracket := \{p\}$$

$$\llbracket q \rrbracket := \{p, q, r, t\}$$

$$\llbracket r \rrbracket := \{r, t\}$$

$$\llbracket t \rrbracket := \{t\}$$

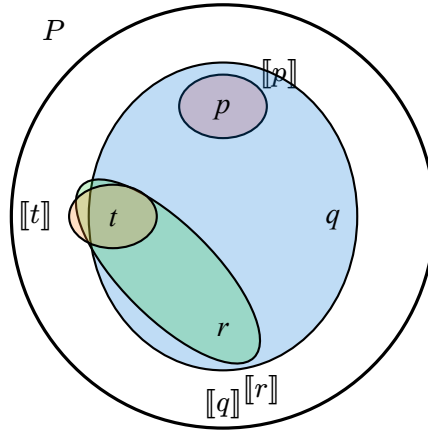


Figure 3: $\mathcal{M}$

$$\mathcal{M} \models \text{All } p \text{ are } p \quad \checkmark$$

$$\mathcal{M} \models \text{All } p \text{ are } q \quad \checkmark$$

$$\mathcal{M} \models \text{All } p \text{ are } r \quad \times$$

$$\mathcal{M} \models \text{All } p \text{ are } t \quad \times$$

$$\mathcal{M} \models \text{All } q \text{ are } p \quad \times$$

$$\mathcal{M} \models \text{All } q \text{ are } q \quad \checkmark$$

$$\mathcal{M} \models \text{All } q \text{ are } r \quad \times$$

$$\mathcal{M} \models \text{All } q \text{ are } t \quad \times$$

$$\mathcal{M} \models \text{All } r \text{ are } p \quad \times$$

$$\mathcal{M} \models \text{All } r \text{ are } q \quad \checkmark$$

$$\mathcal{M} \models \text{All } r \text{ are } r \quad \checkmark$$

$$\mathcal{M} \models \text{All } r \text{ are } t \quad \times$$

$\mathcal{M} \models \text{All } t \text{ are } r \quad \checkmark$

$\mathcal{M} \models \text{All } t \text{ are } p \quad \times$

$\mathcal{M} \models \text{All } t \text{ are } q \quad \checkmark$

$\mathcal{M} \models \text{All } t \text{ are } t \quad \checkmark$

8 Problem 8

Now let's move away from concrete examples and work abstractly. Suppose P is any signature and Γ is any set of P -sentences. Can you *define* the canonical model $\mathcal{M}_\Gamma$? Remember that to define a model you need to specify a domain and, for each symbol in the signature, an interpretation of that symbol as a subset of the domain.

8.1 Solution

$$\forall a \in P, \llbracket a \rrbracket := \{x \in P : \Gamma \vdash \text{All } x \text{ are } a\}$$

9 Problem 9

gain, let P be any signature and Γ be any set of P -sentences. Using your definition of the canonical model $\mathcal{M}_\Gamma$ from the previous problem, rigorously prove that $\mathcal{M}_\Gamma \models \Gamma$; i.e., that $\mathcal{M}_\Gamma$ satisfies every sentence in Γ .

9.1 Solution

$$\mathcal{M}_\Gamma \models \Gamma (\forall \phi \in \Gamma \mathcal{M}_\Gamma \models \phi)$$

Let $\phi \in \Gamma$ so $\exists p, q \in P. \phi = \text{All } p \text{ are } q$

So then $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$.

Let

$$a \in \llbracket p \rrbracket = \{x \in P : \Gamma \vdash \text{All } x \text{ are } a\}$$

So $a \in P$ and $\exists$ derivation $\mathcal{D}$ of $\text{All } a \text{ are } q$

10 Solution 10

Let P be any signature, Γ be any set of P -sentences, and let ϕ be any arbitrary P -sentence. Suppose that $\mathcal{M}_\Gamma \models \phi$. Prove that $\Gamma \vdash \phi$.

10.1 Solution

This is completeness

We know that

$$\mathcal{M}_\Gamma \models \phi$$

This would contain something like

$$\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$$

where

$$\llbracket q \rrbracket = \left\{ x \in P : \Gamma \vdash \underbrace{\text{All } x \text{ are } q}_{\phi} \right\}$$

This then mean thats $\Gamma \vdash \phi$